

SAVEETHA SCHOOL OF ENGINEERING
CSA15- CLOUD COMPUTING AND BIG DATA ANALYTICS

LIST OF EXPERIMENT 5

5) Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS.

SOL

AIM:

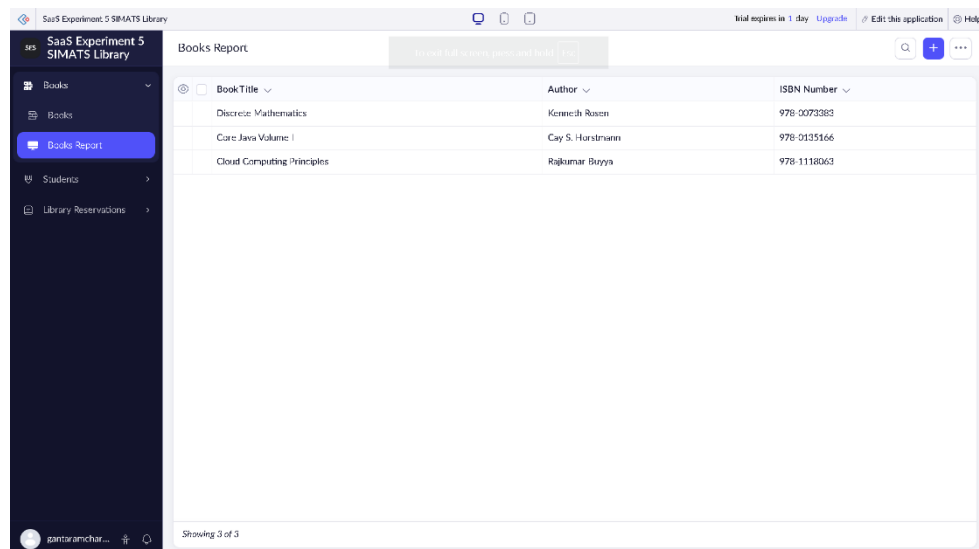
To design and implement a simple cloud-based Library Book Reservation System customized for the SIMATS library using Zoho Creator and deploy it as a Software as a Service (SaaS) application, demonstrating relational table mapping via lookup links, student role handling, and inventory tracking over the internet via a web browser.

PROCEDURE:

- **Sign In:** Sign in to the Zoho Creator cloud platform using the registered student account credentials.
- **Application Shell:** Create a completely fresh standalone application by selecting the **Create from Scratch** option and name it SaaS Experiment 5 SIMATS Library.
- **Form 1 (Books Table):** Design the first data entry form named Books, adding three **Single Line** text fields labeled Book Title, Author, and ISBN Number. Publish and save the form structure.
- **Form 2 (Students Table):** Design the second data entry form named Students to manage member data, adding a **Name** field labeled Student Name and a **Single Line** text field labeled Register Number. Save the form configuration.
- **Form 3 (Library Reservations Relational Table):** Create the third form named Library Reservations to act as the relational junction mechanism connecting students to issued literature:
 - Drag a **Lookup Field** linking directly to the Students form, choosing Student Name as the primary display index.
 - Drag a second **Lookup Field** linking directly to the Books form, choosing Book Title as the primary display index.
 - Add a **Date** field labeled Reservation Date to track library booking timelines.

- **Report Generation:** Automatically generate independent list view reports for Books, Students, and Library Reservations to facilitate multi-tenant search filtering, data browsing, and immediate ledger sorting via the cloud environment.
- **Deployment & SaaS Access:** Click on the blue **Access This Application** button to securely host and deploy the automated library tracking system live onto the Zoho cloud platform.
- **Functional Cloud Testing:** Verify live structural dependencies by sequentially inserting 3 unique sample entries into the Books form, 3 entries into the Students form, and completing cross-table operational lookup mappings inside the dynamic Library Reservations window.

SNAPSHOTS:

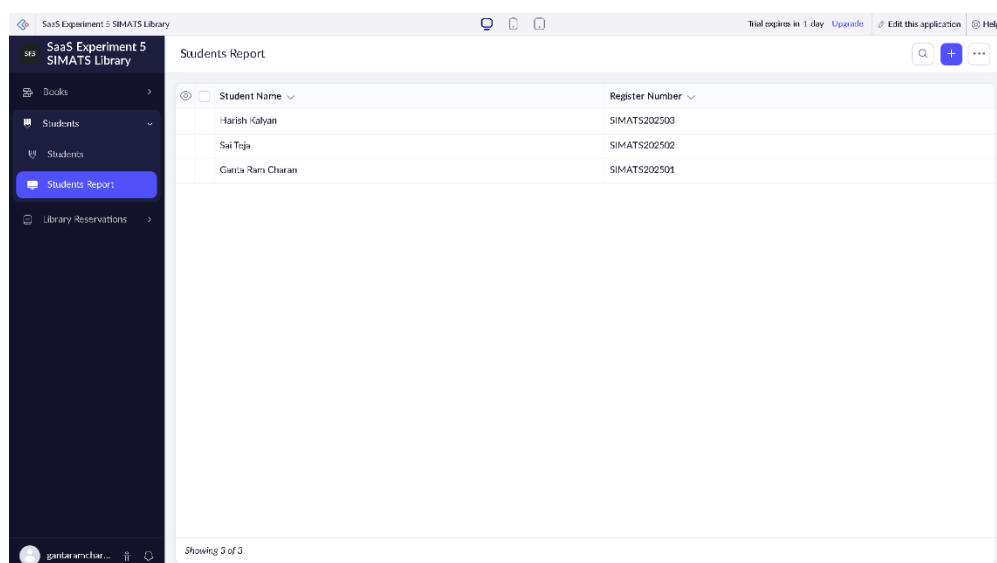


SaaS Experiment 5 SIMATS Library

Books Report

Book Title	Author	ISBN Number
Discrete: Mathematics	Kenneth Rosen	978-0073383
Core Java Volume I	Cay S. Horstmann	978-0135166
Cloud Computing Principles	Rajkumar Buyya	978-1118063

Showing 3 of 3



SaaS Experiment 5 SIMATS Library

Students Report

Student Name	Register Number
Harish Kalyan	SIMATS202503
Sai Teja	SIMATS202502
Ganika Ravi Charan	SIMATS202501

Showing 3 of 3

Select Student	Select Book	Reservation Date
Harish Kalyan	Discrete Mathematics	31-Jul-2026
Sai Teja	Core Java Volume I	30-Jul-2026
Ganta Ram Charan	Cloud Computing Principles	30-Jul-2026

Showing 3 of 3

RESULT:

A relational cloud-based Library Book Reservation System application for the SIMATS library was successfully developed, integrated, and deployed using the Zoho Creator SaaS framework. The architecture enabled robust cross-form lookup connections and immediate multi-user accessibility directly via an internet web browser without local server deployment, successfully fulfilling Software as a Service (SaaS) criterion.